Environment Setup

VirtualBox Setup and Configuration Guide

Overview

Oracle VM VirtualBox is a free, open-source, cross-platform hosted hypervisor that allows you to run multiple guest operating systems (such as Linux, Windows, macOS, or BSD) on a single physical host machine. It is widely used for software testing, development, and secure sandbox environments.

System Requirements & Prerequisites

Before beginning the installation, ensure your host hardware meets the following requirements:

Component	Minimum Requirement	Recommended Specification
Processor	**64-bit x86/AMD64 or ARM64**	**Multi-core with Hardware Virtualization**
Memory (RAM)	**4 GB**	**8 GB or more**
Storage	**30 GB free space**	**50 GB+ per Virtual Machine**
Virtualization	**BIOS/UEFI Disabled**	**VT-x (Intel) or AMD-V (AMD) Enabled**

Operating System ISO: Ensure you have already downloaded the ISO installation image of the guest operating system you plan to run (e.g., Ubuntu, Debian, Windows 11).

Enable Hardware Virtualization (VT-x / AMD-V)

Hardware virtualization must be enabled in your host system's BIOS or UEFI settings to ensure high performance and 64-bit guest support:

1. **Restart your host PC.**
2. **Press the BIOS/UEFI setup key repeatedly during startup (typically F2, F10, F12, or Del).**
3. **Navigate to CPU Configuration, Advanced, or Security.**
4. **Locate Intel Virtualization Technology (VT-x) or AMD-V / SVM Mode and set it to Enabled.**
5. **Save changes and exit BIOS to reboot into your host OS.**

Download and Install VirtualBox

Windows Host:

1. **Go to the official website: virtualbox.org.**
2. **Download the latest installer for Windows hosts.**
3. **Run the downloaded `.exe` installer.**
4. **Accept the default installation components and paths.**
5. **Click Next, proceed through network interface reset warnings, and click Install.**
6. **When prompted by User Account Control (UAC), grant permission.**

Create a New Virtual Machine (VM)

1. **Open Oracle VM VirtualBox Manager.**
2. **Click the New (star icon) button in the top toolbar.**
3. **Name and Operating System:**
    - **Name: Enter a descriptive name, for example `Windows 10 Pro`**
    - **Folder: Choose the destination directory for VM files.**

    - **ISO Image: Browse and select the downloaded guest OS `.iso` file (e.g., `Win10_22H2_English_x64.iso`).**
    - **Check or uncheck Skip Unattended Installation based on whether you prefer automated setup or manual Windows 10 configuration.**
4. **Hardware Allocation:**
    - **Base Memory (RAM): Assign at least 4096 MB (4 GB) for Windows 10 (8 GB recommended for optimal performance), staying within the green zone of your host RAM.**
    - **Processors: Allocate 2 or more CPU cores (at least 2 cores recommended for Windows 10).**
    - **EFI BIOS Setup: Ensure Enable EFI (special OSes only) is checked if installing Windows 10 for modern boot compatibility.**
5. **Virtual Hard Disk:**
    - **Select Create a Virtual Hard Disk Now.**
    - **Set disk size (minimum 50 GB recommended for Windows 10 installations).**
    - **Use dynamic allocation (default) to consume disk space only as files are written.**
6. **Click Finish.**

Configure VM Settings

Select your newly created VM and click Settings (gear icon) to optimize the environment:

- **General > Advanced: Set Shared Clipboard and Drag'n'Drop to Bidirectional.**
- **System > Motherboard: Ensure Enable EFI is enabled if installing a modern OS (required for Windows 11).**
- **Display > Screen: Increase Video Memory to 128 MB and check Enable 3D Acceleration.**
- **Network > Adapter 1:**
    - **Leave set to NAT for simple internet access.**

    - **Switch to Bridged Adapter if the VM needs its own IP address on your local physical network.**

Install the Guest OS

1. **Select the VM and click Start.**
2. **The VM will boot from the mounted ISO image.**
3. **Follow the on-screen installer instructions for the guest operating system (language, partitioning, user account, password).**
4. **When installation finishes, reboot the virtual machine.**
5. **In the VirtualBox menu under Devices > Optical Drives, verify the installation ISO has unmounted.**

Install VirtualBox Guest Additions

Guest Additions provides mouse pointer integration, automatic display resolution scaling, shared folders, and clipboard sharing.

1. **Boot into the guest operating system.**
2. **In the VirtualBox application window menu, click Devices > Insert Guest Additions CD image….**
3. **On Windows Guest: Open File Explorer, double-click the CD drive, and run `VBoxWindowsAdditions.exe`.**
4. **On Linux Guest: Open a terminal and run:**

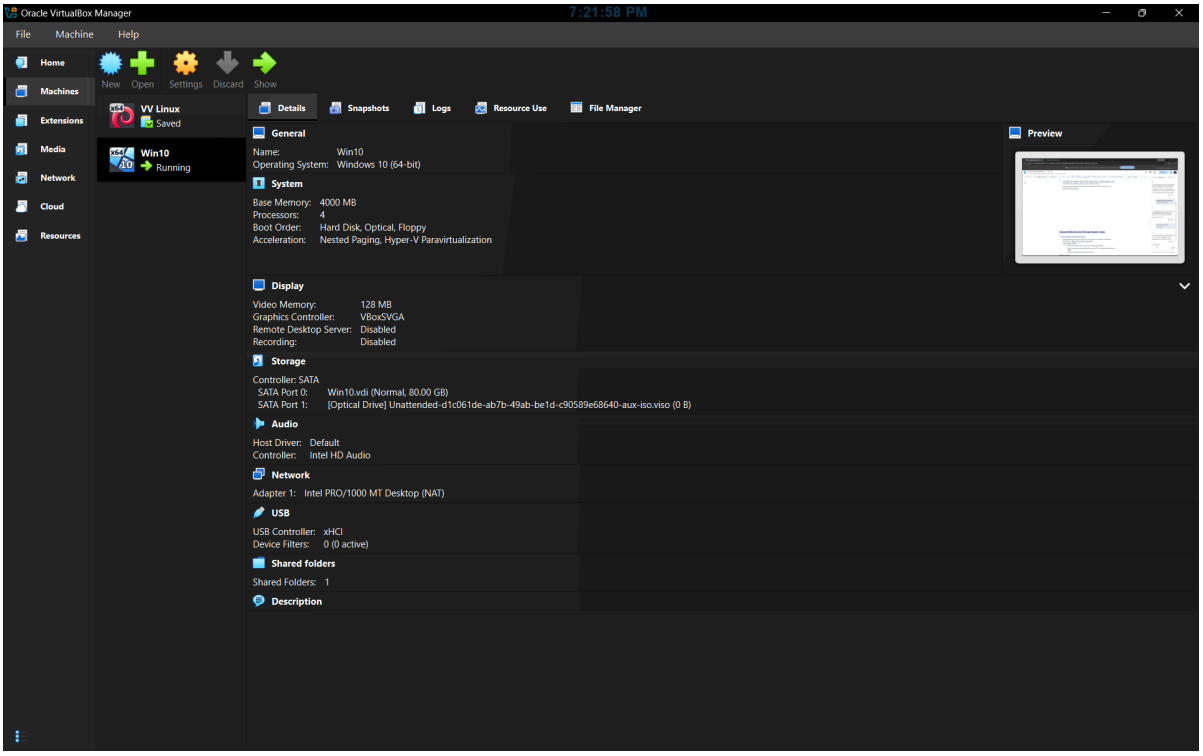
```
sudo apt update
sudo apt install build-essential dkms linux-headers-$(uname -r) -y
sudo /media/$USER/VBox_GAs_*/VBoxLinuxAdditions.run
```

5. **Restart the virtual machine to apply the drivers.**

Best Practices & Troubleshooting

- **Snapshots: Use the Snapshots feature before making major system changes or software updates so you can quickly restore to a clean state.**

- **VT-x/AMD-V Not Available: Verify Windows Hyper-V, WSL2, or Memory Integrity / Core Isolation are not conflicting if using an older VirtualBox release.**
- **Screen Resolution Issues: Verify that Guest Additions is installed and select View > Auto-resize Guest Display.**

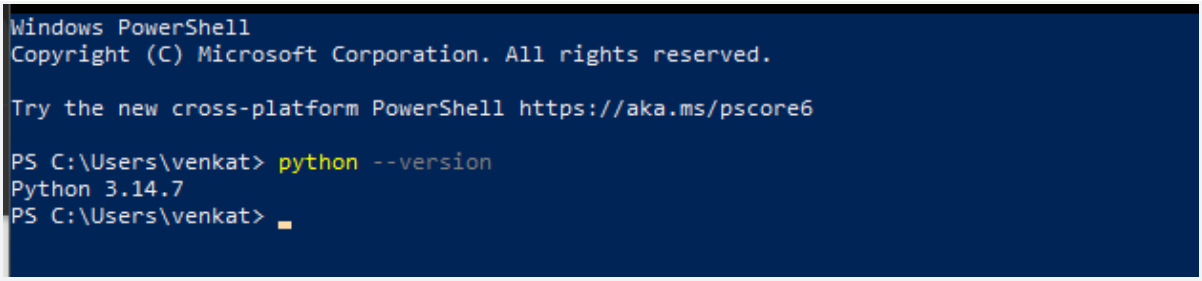


Setup Python3 and Virtual Studio Code

1. Prerequisites & Runtime Setup

- **Operating System: Windows 10 (64-bit) running inside an isolated Virtual Machine (VirtualBox or VMware) or guest host environment.**
- **Install Python Runtime:**
    1. **Download the official Python 3.x (64-bit) installer for Windows.**
    2. **Run the installer and ensure "Add python.exe to PATH" is checked before clicking install.**
    3. **Verify the installation in Windows PowerShell:**

```
python --version
```



2. Workspace & Virtual Environment Configuration

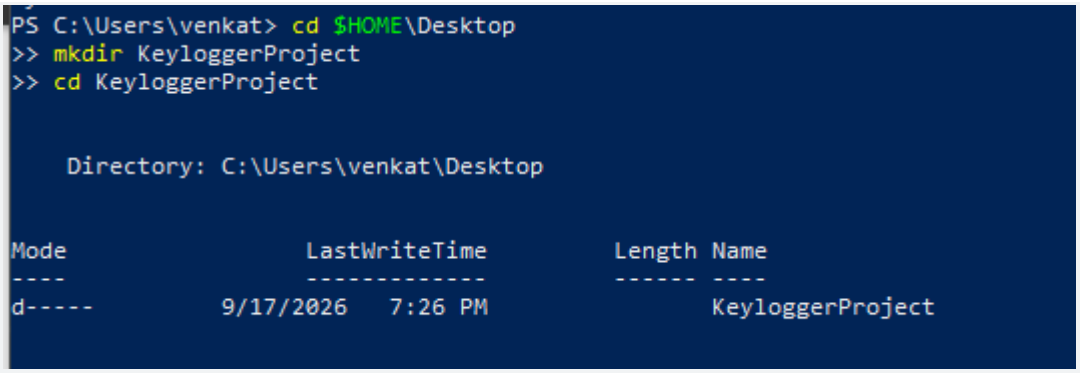
Step 2.1: Project Directory Setup

Open Windows PowerShell (avoid running inside system directories like `C:\Windows\system32`) and create your project directory:

```
cd $HOME\Desktop
```

```
mkdir KeyloggerProject
```

```
cd KeyloggerProject
```



Step 2.2: PowerShell Script Execution Policy

By default, Windows restricts script execution, which blocks virtual environment activation scripts. Bypass this for your active session:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

(When prompted, select [A] Yes to All *).

```
PS C:\Users\venkat\Desktop\KeyloggerProject> Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
>>

Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose you to the
security risks described in the about_Execution_Policies help topic at https:/go.microsoft.com/fwlink/?LinkID=135170. Do you want to
change the execution policy?
[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): A
```

Step 2.3: Virtual Environment (`venv`) Setup & Activation

Isolate your project dependencies from the global system Python installation:

```
# Create the local virtual environment named 'venv'
python -m venv venv
```

```
# Activate the virtual environment
.\venv\Scripts\Activate.ps1
```

Verification: Your PowerShell prompt will now display (venv) at the beginning

```
(venv) PS C:\User\<Username>\Desktop\Keylogger>
```

```
PS C:\Users\venkat\Desktop\KeyloggerProject> python -m venv venv
PS C:\Users\venkat\Desktop\KeyloggerProject>
PS C:\Users\venkat\Desktop\KeyloggerProject>
PS C:\Users\venkat\Desktop\KeyloggerProject>
PS C:\Users\venkat\Desktop\KeyloggerProject> .\venv\Scripts\Activate.ps1
(venv) PS C:\Users\venkat\Desktop\KeyloggerProject>
```

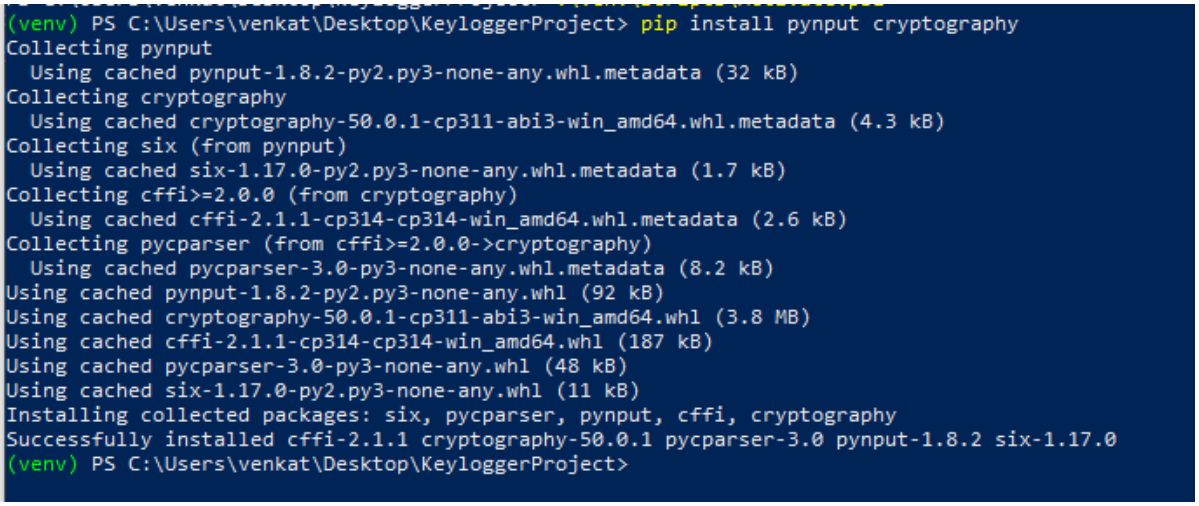
Step 2.4: Dependency Installation

Install the core Python libraries required for input handling and cryptographic storage:

```
pip install pynput cryptography
```

- **pynput: Provides low-level system hooks to asynchronously capture keyboard events.**

- **cryptography: Implements symmetric AES encryption to store captured data securely.**

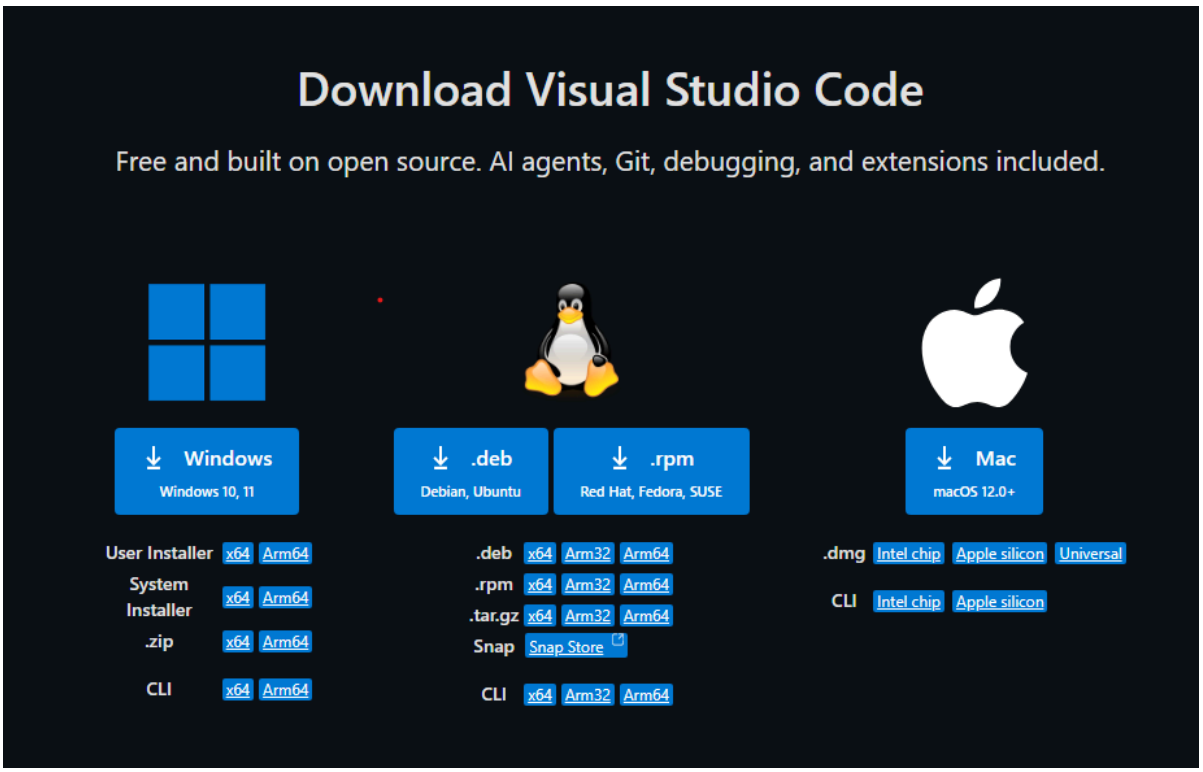


3. Visual Studio Code (VSC) Configuration

Dowload VSC from:

https://code.visualstudio.com/download?_exp_download=fb315fc982

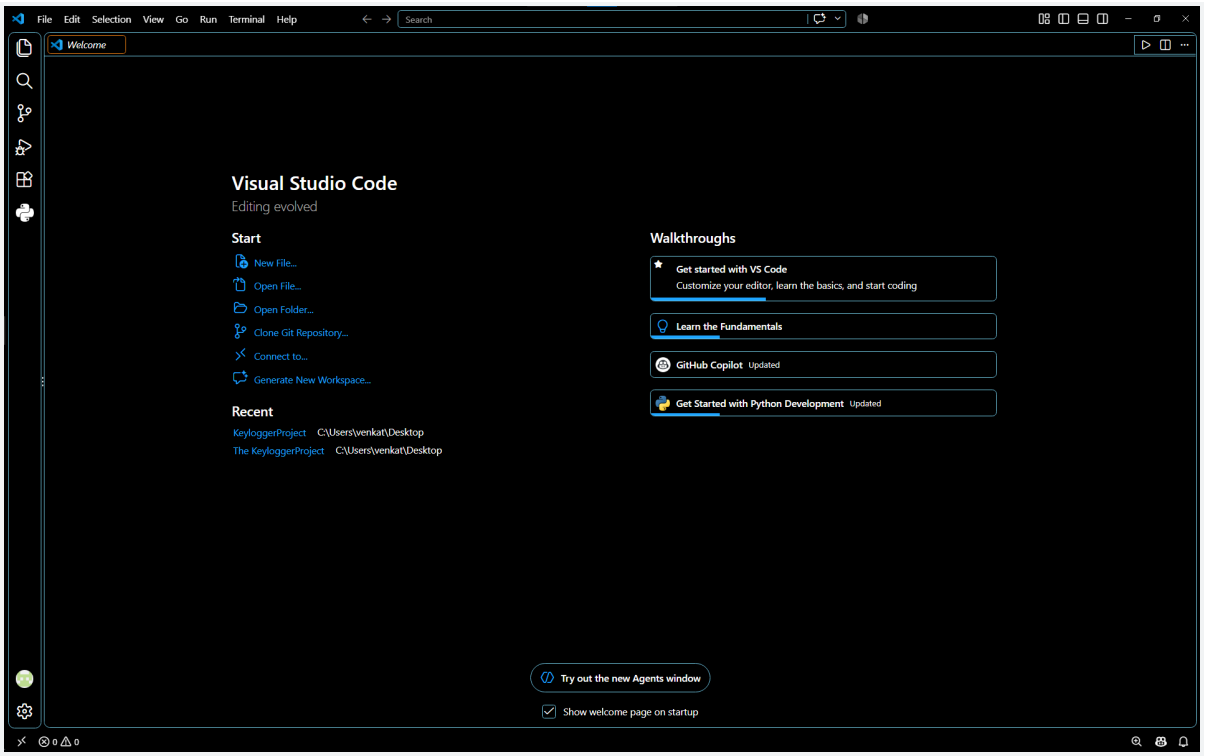
As we are using Windows 10, hit



Step 3.1: Open VS Code in Your Project Workspace

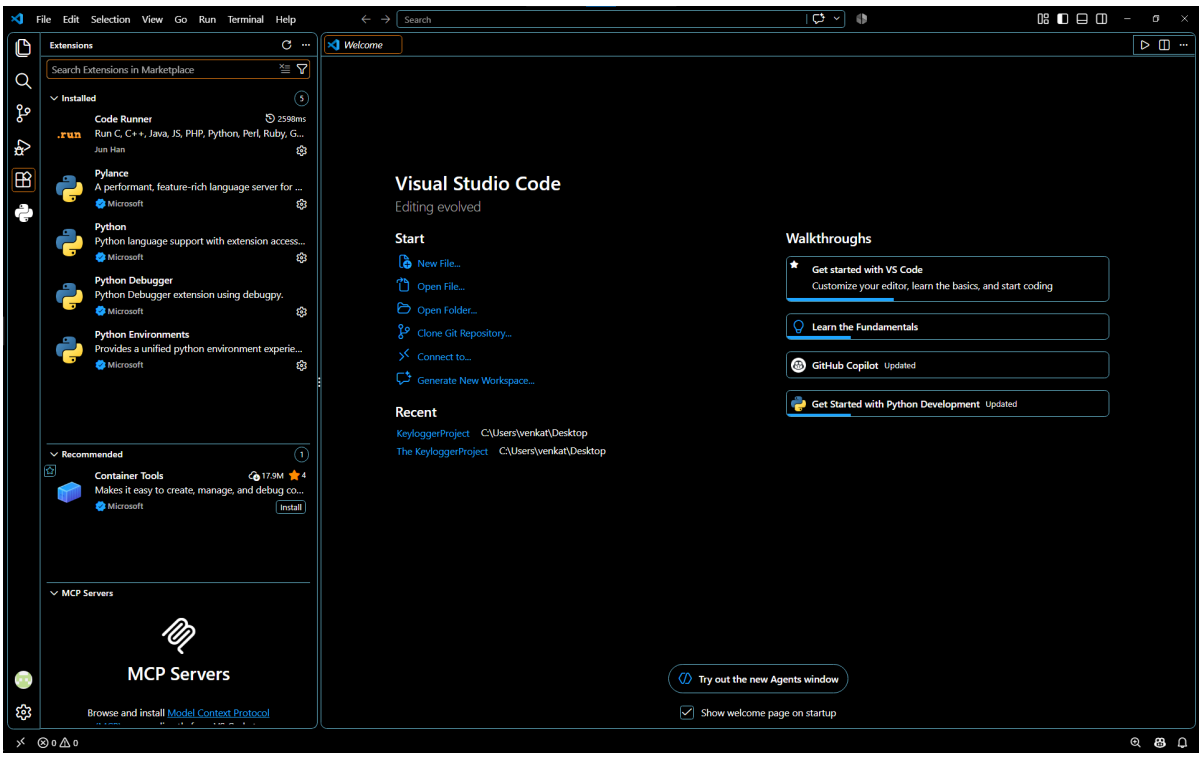
From your activated PowerShell terminal, launch VS Code directly in the workspace directory:

```
code .
```



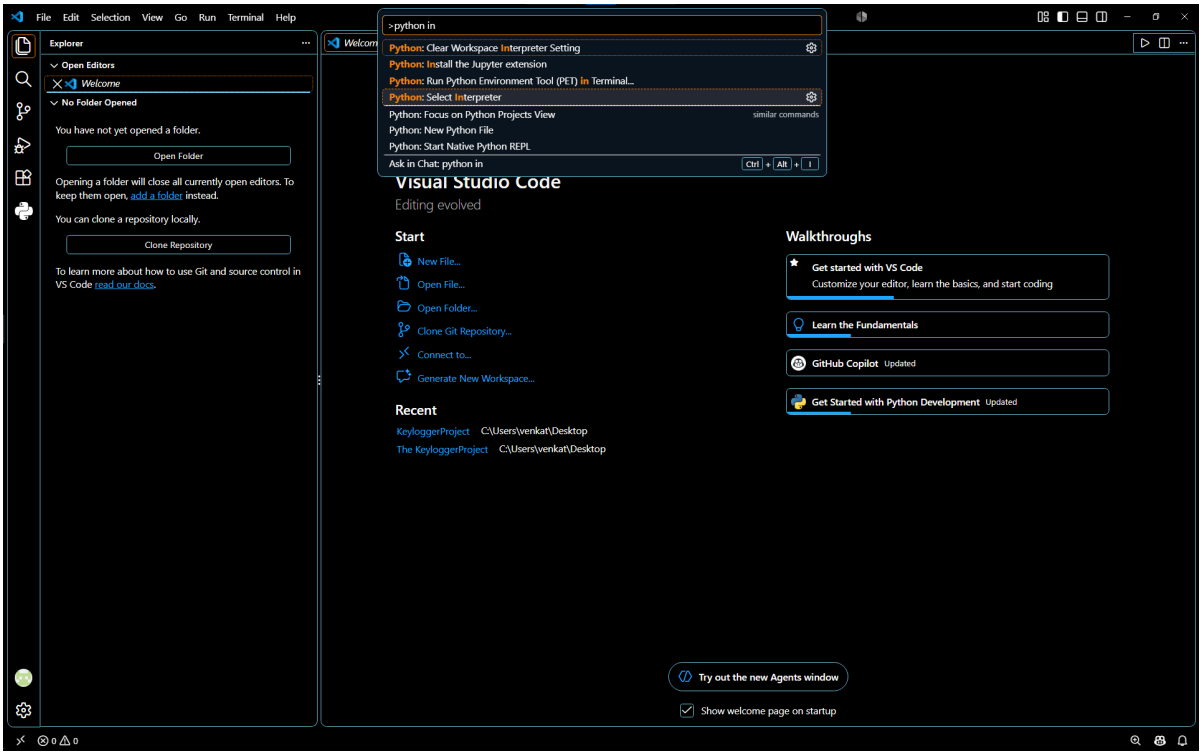
Step 3.2: Install the Python Extension

1. **Inside VS Code, open the Extensions Marketplace (Ctrl + Shift + X).**
2. **Search for and install the official Python extension published by Microsoft.**
3. **Install the necessary extensions which are mentioned in “INSTALLED” section**



Step 3.3: Point VS Code to the Virtual Environment Interpreter

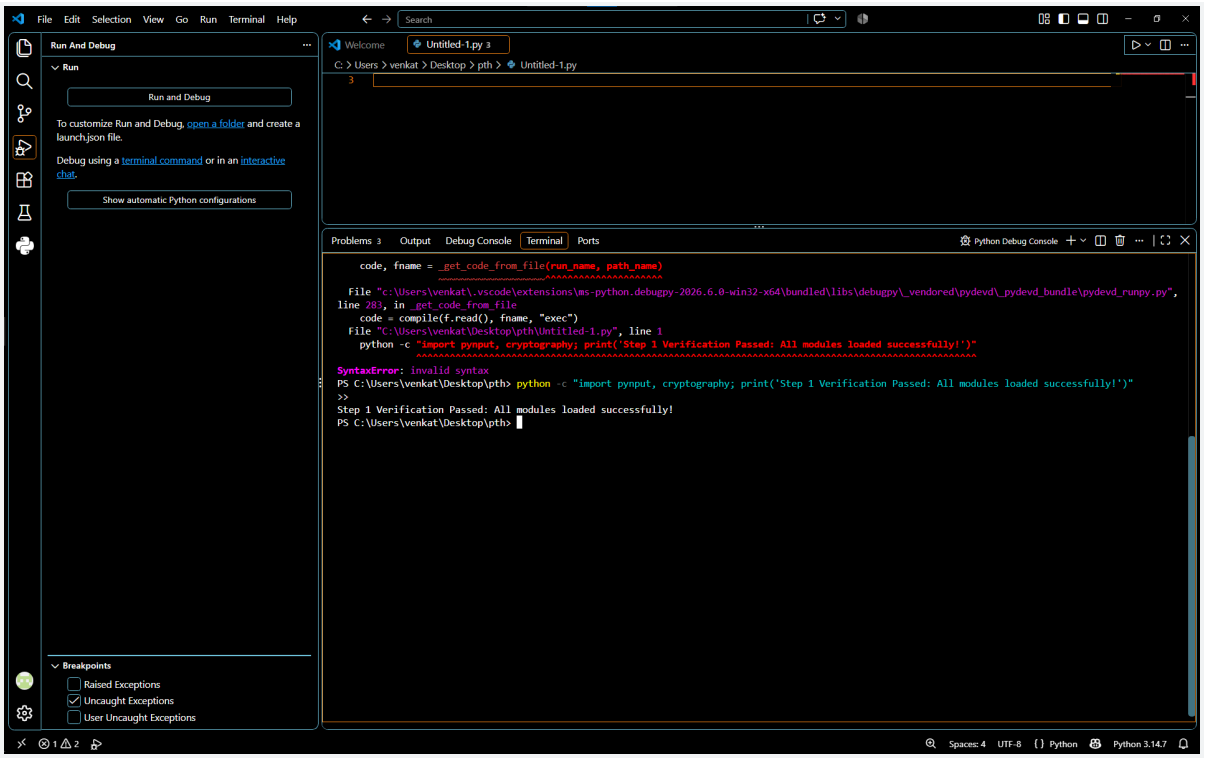
1. **Press Ctrl + Shift + P to open the Command Palette.**
2. **Type Python: Select Interpreter and select it.**
3. **Choose the virtual environment path: .\venv\Scripts\python.exe.**



4. Verification & Sanity Check

To verify that your environment, packages, and paths are configured correctly, run this inline sanity check in your activated terminal:

```
python -c "import pynput, cryptography; print('Step 1 Verification Passed: All modules loaded successfully!')"
```


File Edit Selection View Go Run Terminal Help
Search
Run And Debug
Run
Run and Debug
To customize Run and Debug, open a folder and create a launch.json file.
Debug using a terminal command or in an interactive chat.
Show automatic Python configurations
Breakpoints
Raised Exceptions
Uncaught Exceptions
User Uncaught Exceptions
Welcome
Untitled-1.py 3
C: > Users > venkat > Desktop > pth > Untitled-1.py
3
Problems 3
Output
Debug Console
Terminal
Ports
Python Debug Console
code, fname = _get_code_from_file(run_name, path_name)
File "c:\Users\venkat\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy_vendored\pydevd_pydevd_bundle\pydevd_runpy.py", line 283, in _get_code_from_file
code = compile(f.read(), fname, "exec")
File "C:\Users\venkat\Desktop\pth\Untitled-1.py", line 1
python -c "import pynput, cryptography; print('Step 1 Verification Passed: All modules loaded successfully!')"
SyntaxError: invalid syntax
PS C:\Users\venkat\Desktop\pth> python -c "import pynput, cryptography; print('Step 1 Verification Passed: All modules loaded successfully!')"
>>
Step 1 Verification Passed: All modules loaded successfully!
PS C:\Users\venkat\Desktop\pth>
Spaces: 4
UTF-8
Python
Python 3.14.7